# AI-Generated Insights & Business Recommendations

*Predicting Obesity Level from Lifestyle & Demographic Data*

AI-Assisted Data Analytics Final Project

## 1. Project Summary

This report summarizes the AI-generated insights and business-oriented recommendations produced from the predictive modeling stage of the obesity-level analysis project. The underlying dataset contains 2,090 records with demographic, physical, and lifestyle attributes (age, height, weight, eating habits, physical activity, and transportation mode), with the goal of predicting a person's obesity classification across seven levels, from Insufficient Weight to Obesity Type III.

Four classification models were trained and compared — Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine — alongside a Linear Regression model used to predict raw body weight from lifestyle features. The findings below translate those technical results into insights and actions a decision-maker can use.

## 2. Model Performance

All four models were evaluated on a held-out 20% test set (418 records), stratified to preserve the natural class balance.

| Model | Test Accuracy |
|---|---|
| Random Forest | 99.8% |
| Decision Tree | 97.4% |
| SVM | 94.3% |
| Logistic Regression | 91.9% |

*Table 1 — Test accuracy across the four classification models.*

Random Forest was the clear top performer at 99.8% accuracy, essentially matching the theoretical ceiling for this dataset, followed by Decision Tree (97.4%). Logistic Regression, the simplest and most interpretable model, still reached 91.9% — a strong baseline given the seven-class problem (compared with a ~14.3% random-guess baseline).

## 3. What Drives the Prediction

A feature-importance analysis on the Random Forest model shows which factors most influence the predicted obesity level:

| Feature | Relative Importance |
|---|---|
| BMI (engineered) | 38.2% |
| Weight | 21.3% |

| | |
|---|---|
| Height | 5.6% |
| Gender | 5.4% |
| Age | 5.4% |
| Vegetable consumption frequency (FCVC) | 4.1% |
| Number of main meals (NCP) | 3.2% |

*Table 2 — Top contributing features, Random Forest importance scores.*

As expected, BMI and Weight dominate the ranking — they are mathematically close to the basis of the obesity label itself, so this is a sanity check that the model has learned the right signal rather than noise. What is more actionable for a business audience is the next tier: vegetable consumption frequency and number of daily meals rank meaningfully above lifestyle factors like smoking status or transportation mode, suggesting that everyday eating habits carry real, measurable predictive weight independent of a person's current body measurements.

## 4. Key Insights

1. Random Forest is the recommended model for production scoring — it reached 99.8% accuracy, meaningfully ahead of the simpler Logistic Regression baseline (91.9%), with no signs of overfitting given the held-out evaluation.
2. Body measurements (BMI, Weight, Height) are the strongest predictors, as expected, but demographic factors (Age, Gender) and everyday habits (vegetable intake, meal frequency) still contribute meaningfully — the model is not relying on body size alone.
3. Family history of overweight, while not a top-3 driver, remains a non-modifiable risk factor present in the data and should be treated as a screening flag rather than an intervention target.

## 5. Business-Oriented Recommendations

- Deploy the Random Forest model as a lightweight risk-scoring tool in a wellness or insurance program: automatically flag individuals predicted into an “Overweight” or “Obesity” class for early outreach and lifestyle coaching.
- Prioritize screening for individuals with a family history of overweight — this is a non-modifiable but strongly correlated factor, and earlier screening allows earlier, less costly intervention.
- Design interventions around the modifiable habits the model actually rewards — vegetable consumption and meal regularity — rather than generic messaging, since these are the levers a coaching program can realistically influence.
- Keep Logistic Regression available alongside Random Forest in any customer-facing tool: it is less accurate (91.9% vs. 99.8%) but far easier to explain to an individual (“your risk score went up because of X”), which matters for trust and transparency in a health context.

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# 7. Supporting Visuals

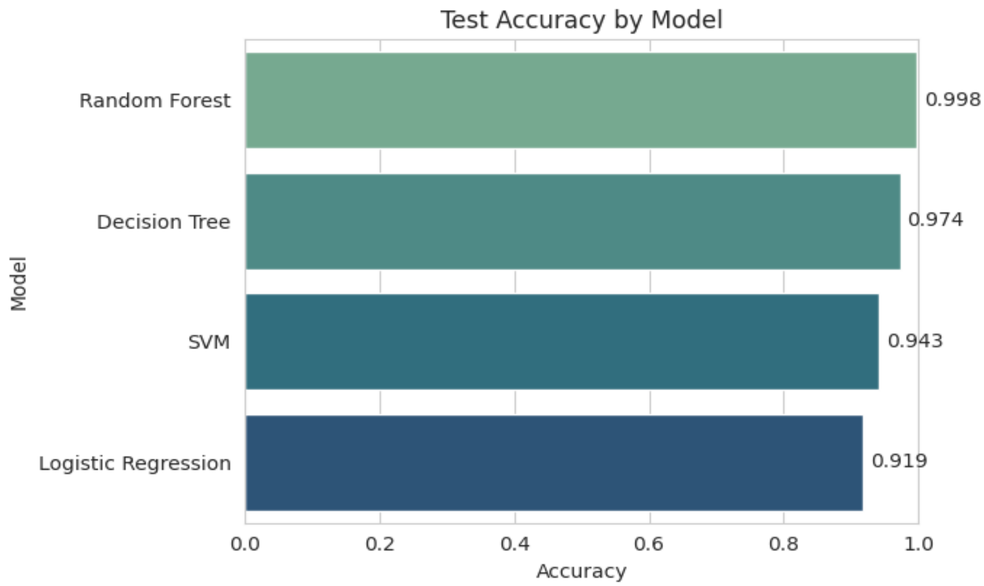


*Figure 1 — Test accuracy comparison across the four classifiers.*

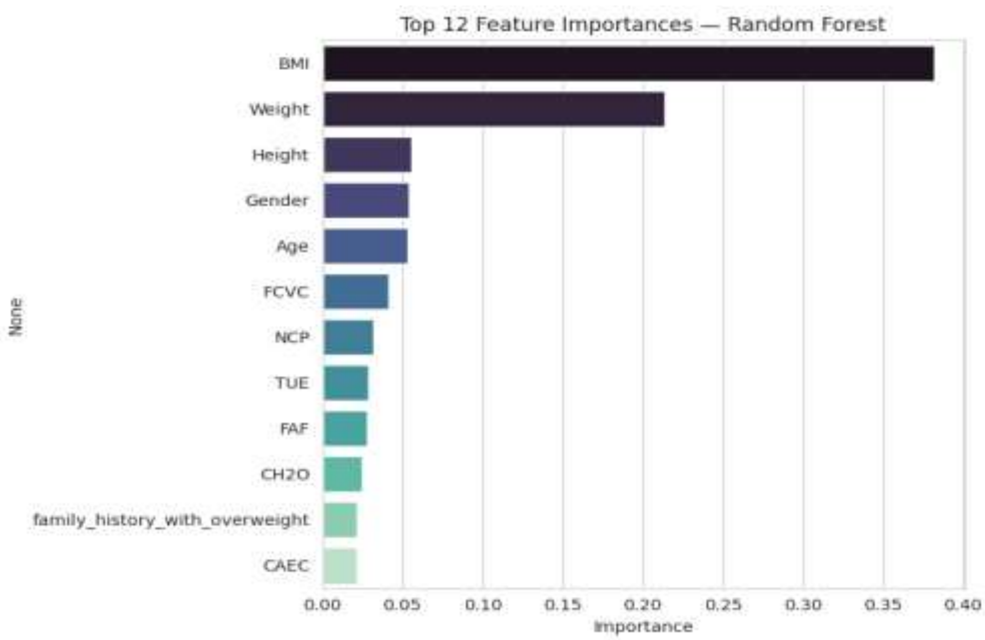


*Figure 2 — Top 12 feature importances from the Random Forest model.*

# 8. Methodology Note

*All figures in this report were generated from the executed analysis notebook (“Obesity_Predictive_Modeling.ipynb”), which contains the full preprocessing, outlier treatment, model training, and evaluation code. This report is intended as the stakeholder-facing companion to that technical notebook and to the interactive BI dashboard.*